

Equations

1 Household

The representative household solves:

$$\max_{\{C_t, I_t, K_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \ln C_t \quad (1)$$

subject to:

$$C_t + I_t + T_t = w_t L_t + r_t K_t + \Pi_t, \quad (2)$$

and the law of motion for capital:

$$K_{t+1} = (1 - \delta)K_t + I_t. \quad (3)$$

The Euler equation is:

$$\frac{1}{C_t} = \beta \frac{1}{C_{t+1}} (1 + r_{t+1} - \delta). \quad (4)$$

2 Final Goods Sectors

There are two final goods sectors, indexed by $j \in \{1, 2\}$. Sector j produces:

$$Y_{j,t} = A_j K_{j,t}^{\alpha_j} L_{j,t}^{\beta_j} (\chi_t E_{j,t})^{1-\alpha_j-\beta_j}. \quad (5)$$

Aggregate output is:

$$Y_t = Y_{1,t} + Y_{2,t}. \quad (6)$$

Sector j chooses capital, labor, and energy services to maximize profits:

$$\Pi_{j,t} = Y_{j,t} - r_t K_{j,t} - w_t L_{j,t} - P_{E,j,t} E_{j,t}. \quad (7)$$

The first-order conditions are:

$$r_t = \alpha_j \frac{Y_{j,t}}{K_{j,t}}, \quad (8)$$

$$w_t = \beta_j \frac{Y_{j,t}}{L_{j,t}}, \quad (9)$$

$$P_{E,j,t} = (1 - \alpha_j - \beta_j) \frac{Y_{j,t}}{E_{j,t}}. \quad (10)$$

Under perfect competition and constant returns to scale:

$$\Pi_{j,t} = 0. \quad (11)$$

Equivalently:

$$Y_{j,t} = r_t K_{j,t} + w_t L_{j,t} + P_{E,j,t} E_{j,t}. \quad (12)$$

Total final-goods-sector profits are:

$$\Pi_t^Y = \sum_{j=1}^2 \Pi_{j,t}. \quad (13)$$

3 Energy Services

Each final goods sector combines fossil and renewable energy into energy services:

$$E_{j,t} = \left[\omega_j E_{f,j,t}^{\frac{\sigma-1}{\sigma}} + (1 - \omega_j) E_{r,j,t}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}. \quad (14)$$

The sector-specific energy price index is:

$$P_{E,j,t} = \left[\omega_j^\sigma P_{f,t}^{1-\sigma} + (1 - \omega_j)^\sigma P_{r,t}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (15)$$

Conditional demand for fossil energy is:

$$E_{f,j,t} = \omega_j^\sigma \left(\frac{P_{f,t}}{P_{E,j,t}} \right)^{-\sigma} E_{j,t}. \quad (16)$$

Conditional demand for renewable energy is:

$$E_{r,j,t} = (1 - \omega_j)^\sigma \left(\frac{P_{r,t}}{P_{E,j,t}} \right)^{-\sigma} E_{j,t}. \quad (17)$$

4 Energy Sector

The energy sector produces fossil and renewable energy using final goods as inputs. Let $M_{f,t}$ and $M_{r,t}$ denote final goods used in fossil and renewable energy production.

Fossil energy production is:

$$E_{f,t} = \frac{z_f}{c_f} M_{f,t}. \quad (18)$$

Renewable energy production is:

$$E_{r,t} = \frac{z_r}{c_r} M_{r,t}. \quad (19)$$

Equivalently, the final-goods inputs required for energy production are:

$$M_{f,t} = \frac{c_f}{z_f} E_{f,t}, \quad M_{r,t} = \frac{c_r}{z_r} E_{r,t}. \quad (20)$$

Under perfect competition, fossil and renewable energy prices are equal to their marginal private costs:

$$P_{f,t} = \frac{c_f}{z_f}, \quad (21)$$

$$P_{r,t} = \frac{c_r}{z_r} - s_t. \quad (22)$$

The subsidy is assumed to satisfy:

$$0 \leq s_t < \frac{c_r}{z_r}. \quad (23)$$

Energy-sector profits are:

$$\Pi_t^E = P_{f,t} E_{f,t} + P_{r,t} E_{r,t} - M_{f,t} - M_{r,t} + s_t E_{r,t}. \quad (24)$$

Using the pricing equations and perfect competition:

$$\Pi_t^E = 0. \quad (25)$$

Total profits rebated to households are:

$$\Pi_t = \Pi_t^Y + \Pi_t^E. \quad (26)$$

5 Government

The government subsidizes renewable energy production. The subsidy is financed by a lump-sum tax on households:

$$T_t = s_t E_{r,t}. \quad (27)$$

6 Market Clearing

Goods market clearing is:

$$Y_t = C_t + I_t + M_{f,t} + M_{r,t}. \quad (28)$$

Using the energy production technologies:

$$Y_t = C_t + I_t + \frac{c_f}{z_f} E_{f,t} + \frac{c_r}{z_r} E_{r,t}. \quad (29)$$

Capital market clearing is:

$$K_t = K_{1,t} + K_{2,t}. \quad (30)$$

Labor market clearing is:

$$L_t = L_{1,t} + L_{2,t}. \quad (31)$$

Fossil energy market clearing is:

$$E_{f,t} = E_{f,1,t} + E_{f,2,t}. \quad (32)$$

Renewable energy market clearing is:

$$E_{r,t} = E_{r,1,t} + E_{r,2,t}. \quad (33)$$

7 Competitive Equilibrium

A competitive equilibrium is a sequence of quantities:

$$\{C_t, I_t, K_t, Y_t, Y_{1,t}, Y_{2,t}, E_{j,t}, E_{f,j,t}, E_{r,j,t}, E_{f,t}, E_{r,t}, M_{f,t}, M_{r,t}\}, \quad t = 0, 1, 2, \dots \quad (34)$$

and prices:

$$\{w_t, r_t, P_{f,t}, P_{r,t}, P_{E,j,t}\}, \quad t = 0, 1, 2, \dots \quad (35)$$

such that:

1. the household satisfies its budget constraint, capital accumulation equation, and Euler equation;
2. final goods firms satisfy their first-order conditions;
3. final goods firms minimize the cost of energy services;
4. the energy sector satisfies its production and pricing equations;
5. the government budget constraint holds;
6. goods, capital, labor, fossil energy, and renewable energy markets clear.

8 Steady State

In steady state, all variables are constant over time. The Euler equation implies:

$$r = \frac{1}{\beta} - 1 + \delta. \quad (36)$$

Investment satisfies:

$$I = \delta K. \quad (37)$$

The steady-state resource constraint is:

$$Y = C + \delta K + \frac{c_f}{z_f} E_f + \frac{c_r}{z_r} E_r. \quad (38)$$